ASHUTOSH SRIVASTAVA

SENIOR AI ENGINEER | FULL STACK ENGINEER

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PROFESSIONAL SUMMARY

Senior AI Engineer and Full Stack Engineer with ~8 years of progressive experience architecting, building, and deploying enterprise-grade software products, distributed backend services, and production Generative AI platforms. Proven expertise across the complete AI lifecycle, specializing in **Azure AI Foundry, Autonomous Reliability Multi-Agent Architectures, production RAG pipelines, AI Evaluation & Quality Benchmarking, and AI Red Teaming** across Energy & Utilities, Financial Services, LegalTech, Automotive, and Omnichannel Retail domains.

Demonstrated track record of taking complex AI applications from end-to-end architecture to production deployment—achieving 95%+ document extraction precision and substantial cost savings for Fortune 500 enterprises (including Shell, General Motors, EY, and Legal & General). Deep hands-on proficiency in **Python, LangChain, LangGraph, LlamaIndex, Azure OpenAI (GPT-4o), Azure AI Search, Microsoft Fabric, Golang, Node.js, React, Docker, Kubernetes, AWS, and Microsoft Azure**. Expert in high-concurrency microservices, real-time WebSocket audio streaming, CI/CD automation, and operations monitoring.

CORE SKILLS & TECHNICAL PROFICIENCY

Generative AI & Agentic AI:	Azure AI Foundry, Large Language Models (LLMs), Agentic AI Systems, Multi-Agent Architectures (LangGraph, LangChain), Autonomous Reliability Multi-Agent Workflows, Production RAG Pipelines, LlamaIndex, AI & LLM Evaluation Frameworks (Groundedness, Relevance, Completeness), AI Red Teaming (Adversarial Robustness & Jailbreak Defense), Prompt Engineering (Chain-of-Thought), Tool/Function Calling, Structured Outputs (JSON Schema), Microsoft Copilot Extensibility
Retrieval, Search & NLP:	Azure AI Search / Cognitive Search, Hybrid Search (Dense Vector + Sparse/BM25), Semantic Reranking, Vector Embeddings, Pinecone, Document Chunking & Indexing Strategies, Azure Document Intelligence, Natural Language Processing (NLP), Pandas, NumPy
Programming & Backend:	Python, Golang, JavaScript, TypeScript, SQL, Node.js, FastAPI, Express.js, Restify, Sequelize, Microservices Architecture, RESTful APIs, Real-Time WebSockets (Socket.io), Multi-threaded Ingestion Mechanics
Frontend & UI Engineering:	React.js, Redux, Angular, HTML5, CSS3, Operations Dashboards, Responsive UI/UX Design, State Management, Third-Party API Integrations, Full-Stack Web Applications
Cloud & Data Engineering:	Microsoft Azure (OpenAI, AI Foundry, Functions, Event Hubs, Key Vault, Managed Identity, Blob Storage, Application Insights), AWS (Lambda, API Gateway, SQS, Kinesis, Kinesis Analytics, Athena, Glue, S3, DynamoDB, CloudFront), Microsoft Fabric Pipelines, Serverless Data Lake ETL
DevOps, Containers & Databases:	Docker, Kubernetes (K8s), Container Orchestration, CI/CD Pipelines (GitHub Actions / Azure DevOps), Automated Deployment Workflows, MongoDB, PostgreSQL, MySQL, Redis, InfluxDB, Telegraf, Grafana, Datadog, Git, Linux
Enterprise Domains:	Energy & Petrochemicals (Reliability Engineering, CMMS / SAP PM, FMEA/FMECA, FLOC), Governance, Risk & Compliance (GRC), Financial Document Review & Audit Tech, LegalTech Contract Intelligence, Automotive Decision Support, Omnichannel Retail Platforms

PROFESSIONAL EXPERIENCE

IBM | *Senior Application Consultant / AI Engineer* **Nov 2020 – Present | Bengaluru, India**

Shell (Client Location) — Enterprise RAG, Agent API Platform & Reliability Multi-Agent System (Azure AI Foundry) *(Apr 2026 – Present)*

- **Architected and Deployed an Autonomous Reliability Multi-Agent System:** Engineered a coordinated multi-agent workflow to automate equipment health diagnostics, analyze failure modes (FMEA/FMECA), and evaluate Functional Locations (FLOC) by orchestrating specialized domain agents.
- **Integrated CMMS & Work Order Automation:** Interfaced agentic workflows with enterprise Computerized Maintenance Management Systems (CMMS/SAP PM), automating root-cause insights, dynamic mitigation plans, and preventive maintenance task recommendations for maintenance planners and reliability engineers.
- **Built Enterprise Agent API & RAG Architecture:** Architected and deployed an end-to-end, production-grade Agent API platform and RAG pipeline using Azure AI Foundry, empowering Subject Matter Experts (SMEs) with rapid, high-precision technical knowledge discovery across complex engineering artifacts.
- **Engineered High-Precision Document Ingestion:** Built scalable ingestion pipelines covering multi-format technical manual parsing, semantic chunking, and high-dimensional vector embeddings, achieving 95%+ retrieval accuracy and extraction precision.
- **Automated AI Evaluation & Observability:** Implemented an enterprise-grade AI evaluation framework to benchmark groundedness, context relevance, answer completeness, and hallucination metrics across live maintenance and engineering queries.
- **Hardened Security & CI/CD Operations:** Conducted comprehensive AI Red Teaming and vulnerability assessments to defend against prompt injection and data exfiltration; automated CI/CD deployment pipelines and built real-time telemetry dashboards tracking latency, token usage, and system health.

Environment: Azure AI Foundry, Azure OpenAI (GPT-4o), LangGraph, LangChain, Python, CMMS / SAP PM APIs, Azure AI Search, Docker, Vector DBs, REST APIs, CI/CD.

Legal & General (Microsoft Client) — Agentic Voice & Knowledge Search Platform *(Nov 2025 – Mar 2026)*

- Architected an enterprise Agentic AI conversation intelligence platform using Python, Azure OpenAI, LangChain, and Azure AI Search for live customer-support operations.
- Engineered real-time audio streaming ingestion pipelines over WebSockets integrated with Azure Speech Services for sub-second speech-to-text transcription.
- Implemented autonomous multi-agent workflows executing automated intent classification, call summarization, disposition tagging, and Next Best Action (NBA) CRM case automation.
- Built event-driven, high-concurrency microservices using Azure Event Hubs and serverless Azure Functions to process high-volume live conversation telemetry.

Halliburton (Microsoft Client) — Enterprise Copilot & Microsoft Fabric Data Pipelines *(Aug 2025 – Nov 2025)*

- Engineered enterprise AI capabilities leveraging Microsoft Copilot Extensibility, integrating intelligent AI workflows into internal enterprise applications.
- Designed and implemented Azure AI Search architecture for indexing and contextual retrieval of enterprise operational data and unstructured engineering knowledge.
- Built high-throughput distributed data ingestion pipelines using Microsoft Fabric and Python multi-threaded processing mechanisms, accelerating large-scale document indexing throughput.

Ballard Spahr — AI Legal Communications Analysis & Triage *(Jun 2025 – Aug 2025)*

- Developed an AI-powered legal email triage pipeline using Large Language Models and NLP to automatically evaluate incoming correspondence for priority, importance, and required actions.
- Engineered structured entity and metadata extraction pipelines from legal communications while adhering to rigorous legal compliance and privacy standards.

Ernst & Young (EY - AIFS) — AI-Powered Financial Document Review Platform *(Jan 2024 – Mar 2024 & Feb 2025 – Jun 2025)*

- Designed and implemented secure RESTful APIs using FastAPI and Python for the complete financial document processing lifecycle (upload, rule execution, and insights retrieval).
- Implemented advanced Chain-of-Thought (CoT) prompt engineering strategies and JSON schema validations, minimizing model hallucinations and enforcing 100% compliance with financial rules.
- Engineered custom chunking, metadata extraction, and indexing logic using Azure AI Search and Document Intelligence, securing services via Azure Managed Identity and Key Vault.

Link Logistics (Blackstone) — Legal Contract Abstraction & RAG Platform *(Nov 2024 – Feb 2025)*

- Developed and optimized production Retrieval-Augmented Generation (RAG) workflows using LlamaIndex and Azure OpenAI for automated legal contract abstraction.
- Built scalable document ingestion and processing pipelines handling high-volume commercial real estate leases, generating structured summaries and key clause extractions.

General Motors (GM - Microsoft Client) — Focused Multi-Agent Architecture & LangGraph *(Sep 2024 – Nov 2024)*

- Architected a highly focused, deterministic Multi-Agent System using **LangGraph state machines and LangChain** to coordinate autonomous workflows across distributed enterprise automotive domains.
- Designed a hierarchical agent topology with a master supervisor agent delegating specialized sub-tasks to dedicated agents for multi-source knowledge retrieval, factual cross-verification, reasoning, and structured synthesis.
- Engineered cyclic graph execution, stateful memory persistence, and dynamic tool-routing mechanisms, preventing infinite loops and ensuring resilient task execution.
- Integrated Python-based multi-agent backends with Azure Bot Services and Azure Cognitive Search to deliver context-aware conversational intelligence for GM engineering teams.

Con Edison & Southern California Edison (Edison GPT) — Utility GRC AI Chatbots *(Aug 2023 – Aug 2024)*

- Architected enterprise AI chatbots for Con Edison's Governance, Risk, and Compliance (GRC) knowledgebase using Python, Node.js, React, Azure OpenAI, and Azure Cognitive Search.
- Implemented dynamic RAG pipelines integrating utility whitepapers, policy materials, tone violation checks, and key compliance terms.

Three Stripes (AI Product Development) — GenAI DevOps Productivity Suite *(Mar 2023 – Aug 2023)*

- Contributed to building a foundational GenAI productivity platform leveraging Azure OpenAI foundation models to streamline development, testing, and QA workflows.

CliftonLarsonAllen (CLA) — Audit & Financial Microservices *(Feb 2023 – Mar 2023)*

- Developed and optimized backend microservices and REST APIs using Node.js, MongoDB, Docker, and PostgreSQL supporting audit, tax, and consulting platforms.

Where to Buy (WTB) - PriceSpider — Distributed Commerce Microservices *(Nov 2020 – Jan 2023)*

- Engineered and maintained distributed high-concurrency microservices (locate, wtb-api, data pipeline) in Golang and Node.js for a global retail commerce platform.
- Deployed containerized services to AWS using Docker and Kubernetes (K8s), ensuring high availability across high-traffic consumer brand portals.
- Implemented robust monitoring and observability stacks using Telegraf, InfluxDB, and Grafana to track API latency, throughput, and service health; built UI analytics in React.

ACE TURTLE SERVICES LTD | *Software Developer / Product Developer* — Feb 2020 – Nov 2020 | Bengaluru, India

Rubicon 3 — Omnichannel Retail Enterprise Platform & AWS Data Lake

- Developed core application frameworks and microservices using Node.js, Express, Restify, and MongoDB to manage distributed inventory, product catalogs, orders, and logistics.
- Architected and deployed serverless data lake and analytics infrastructure on AWS using AWS Lambda, API Gateway, SQS, AWS Glue, Athena, S3, and DynamoDB for large-scale ETL processing.
- Engineered real-time data streaming pipelines leveraging AWS Kinesis and Kinesis Analytics; built responsive administrative interfaces in Angular.

AUCTIONSOFTWARE.COM (DevelopScripts LLC) | *Software Development Engineer (Full-Stack)* — Aug 2018 – Feb 2020 | Chennai, India

Enterprise E-Commerce, Live Bidding Platforms & Warehouse Management Systems

- **Mazzady & Auction E-Commerce Engines:** Architected and developed scalable bidding platforms supporting Forward, Reverse, and Penny auction paradigms using Node.js, Express, Sequelize, MySQL, and React; integrated secure multi-gateway payment processing with Stripe and PayPal.
- **Sea Dwelling Warehouse Management System:** Built an end-to-end warehouse and order lifecycle application from scratch using Node.js, React, MySQL, and Sequelize; engineered real-time communication modules via WebSockets (Socket.io), REST APIs for iOS clients, and automated shipping logistics via ShipStation.
- **BlazeBite & Ideal Floor Full-Stack Applications:** Developed online food ordering platform modules using Node.js, Angular, MongoDB, Docker, and Stripe; executed complete platform migration of interior design e-commerce systems from Magento to MERN stack (MongoDB, Express, React, Redux, Node.js).

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | Bharath Institute of Higher Education and Research, Chennai, India (2014 – 2018) | **Grade A (First Class)**
Senior Secondary (Class XII): PCM 89% | Overall 82% (2014)
Database Developer Project & Research: Indian Institute of Technology (IIT), Madras (2016 – 2017)
Key Certifications: Azure AI Foundry & Microsoft AI Ecosystem • Node.js Design Patterns (LinkedIn) • Exploring & Mastering the Full Go Featureset (LinkedIn) • Learning AWS for Developers • Docker Essential Training & Kubernetes • React & Advanced TypeScript • Java Data Structures & Algorithms (Pluralsight)